**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 117 · Lesson 117 · Waterfall and Agile Development

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the difference between linker and loader?

Q2. What is the time complexity of sequential search in the worst case?

Q3. After pushing A then B onto a stack, which is popped first?

Q4. Why must a binary search operate on ordered data?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

waterfall — Stage-based plan-driven development lifecycle.

agile — Iterative and adaptive development principles.

iteration — Repeated short cycle yielding feedback/increment.

requirements — Functions/constraints a system must satisfy.

## New learning

Waterfall lifecycle — Sequential analysis, design, implementation, testing and maintenance.

Agile methods — Iterative delivery, feedback and reprioritisation in short cycles.

Choose on evidence — Stability, regulation, stakeholder access and change risk affect suitability.

## Worked example

An air-traffic subsystem has carefully fixed specifications and requires formal approvals. Select a development approach.

1. Model answer — A plan-driven/waterfall approach can support documented phase gates and traceability; however, testing and validation must still occur throughout as appropriate. Safety assurance matters more than simply labelling it waterfall. — Check: link each claim to the scenario and use precise OCR terminology.

# We Do · Practise with guidance

A community app changes weekly after users test prototypes. Which approach is more suitable?

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Agile iterations with prioritised backlog, frequent releases and real user feedback. A strictly sequential plan risks discovering changed needs late. Maintain security and test discipline.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Describe the waterfall stages and one drawback of late requirement changes. [6]

### Question 2

Explain two features of agile development in a public-services app. [6]

### Question 3

Evaluate waterfall versus agile for replacing an exam-management system subject to fixed deadlines and changing rules. [8]

# Exit Ticket · Answer before checking

Agile means a project needs no testing or documentation.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Linker resolves modules/symbols; loader maps executable into memory to run.

Q2. O(n).

Q3. B, because a stack is LIFO.

Q4. It discards half the search space using an ordering comparison.

### You Do pages · Independent questions

Q1. Clear ordered stages [4]; late changes cause costly rework/approval delays [2].

Q2. Iterative increments [2], stakeholder feedback [2], adaptive backlog/prioritisation or continuous tests [2].

Q3. Fixed controls/documentation [2], agility for regulatory change [2], test and migration risks [2], justified hybrid/choice [2].

### Exit Ticket · Model answer

False. Quality activities remain essential.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**